



HeeraMon

Matching & Preference Specification

Matching Specification Document

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1 Purpose

This document defines how HeeraMon represents user preferences and determines how well a job fits those preferences.

The goal is not to produce a generic job-ranking score. The goal is to determine whether an opportunity is worth the user's attention and why.

2 Preference Model

User preferences should be represented as structured records derived from natural-language conversation.

Each preference should contain, where applicable:

- Preference type
- Value
- Priority
- Source
- Confidence
- Status
- Timestamp

Example:

```
1 Category: Salary
2 Value: 40,000 BDT
3 Type: Hard
4 Source: User explicitly stated
5 Confidence: High
```

3 Preference Types

HeeraMon uses four important classifications:

3.1 Hard Requirements

Conditions that normally must be satisfied.

Examples:

- Minimum salary
- Remote-only requirement
- Required location
- Required employment type

A job violating a hard requirement should normally not be presented as a normal match.

3.2 Soft Preferences

Desirable conditions that can be traded off.

Examples:

- Short commute
- Flexible hours
- Lunch benefits
- Preferred technology stack

A job can remain relevant while failing one or more soft preferences.

3.3 Learned Preferences

Preferences inferred from repeated user behavior or previous decisions.

Example: User repeatedly rejects jobs below 40k.

HeeraMon may infer that salary is important, but should ask for confirmation before permanently treating the inferred threshold as a hard requirement.

3.4 Unknown

Information that has not been established.

Unknown must never be treated as either satisfied or violated without evidence.

4 Preference Priority

When preferences conflict, HeeraMon should generally consider them in this order:

**Explicit Hard Requirement → Confirmed Preference → Soft Preference →
Inferred Preference**

However, the agent should ask the user when a conflict materially affects the recommendation.

Temporary conversational requests should not automatically become permanent preferences.

5 Job Matching

Matching should consider both constraint satisfaction and overall fit.

For each job, HeeraMon should determine:

- Hard requirements satisfied?

- Soft preferences satisfied?
- Important preferences unknown?
- Major conflicts?
- Relevant trade-offs?
- Confidence in the assessment?

A useful internal representation is:

```

1 Job
2     Hard constraints
3         Satisfied
4         Violated
5         Unknown
6     Soft preferences
7         Satisfied
8         Not satisfied
9         Unknown
10    Trade-offs
11    Relevant differences

```

6 Matching Logic

Matching should follow a staged approach:

1. **Eligibility check** — Identify violations of explicit hard requirements.
2. **Preference comparison** — Compare remaining jobs against soft and confirmed preferences.
3. **Trade-off identification** — Identify meaningful advantages and disadvantages.
4. **Uncertainty assessment** — Identify important missing or unreliable information.
5. **Fit explanation** — Generate a concise explanation of why the job is or is not worth considering.

The system may use numerical scores internally where useful, but a score must not replace the underlying reasoning or explanation.

7 Trade-Off Handling

HeeraMon should not blindly reject every imperfect job.

Example:

```

1 User preference:
2 Salary      40k
3 Commute     30 minutes
4 AI/ML role
5 7-hour workday preferred

```

```
6
7 Job:
8 Salary: 45k
9 Commute: ~60 minutes
10 AI/ML: Yes
11 Workday: 8 hours
```

HeeraMon should explain:

Salary and role match your requirements, but the commute and working hours are worse than your preferences.

The user can then decide whether the trade-off is acceptable.

If the violated condition is explicitly hard, the job should normally be excluded unless the user has allowed exceptions.

8 Learning From Decisions

User actions provide evidence about preferences.

Examples:

- Repeatedly rejecting low-salary jobs
- Consistently accepting remote roles
- Rejecting long commutes
- Asking repeatedly about work-life balance

HeeraMon may use these patterns to improve future matching.

However:

Observed behavior $\neq$ confirmed requirement.

When an inferred preference becomes important enough to materially change future results, HeeraMon should request confirmation.

9 Explainability

Every meaningful recommendation should be explainable in human terms.

The user should be able to understand:

- Why this job was considered.
- What matches their preferences.
- What does not match.
- What information is unknown.
- What trade-offs exist.

- Which preference influenced the decision.

HeeraMon should avoid presenting unexplained numbers such as:

“Match score: 87%”

without providing the underlying reasoning.

10 Boundaries

The initial release does not require:

- sophisticated machine-learning recommendation models;
- complex behavioral prediction;
- embeddings-based ranking everywhere;
- large-scale personalization models;
- perfect commute calculations;
- automated estimation of unknown job attributes.

Deterministic filtering combined with structured preference comparison and LLM-assisted interpretation is sufficient.

The architecture should allow more advanced ranking and personalization to be added later.

11 Success Criteria

The matching system is successful when HeeraMon can:

1. Convert natural-language preferences into structured requirements.
2. Distinguish hard, soft, learned, and unknown preferences.
3. Exclude or appropriately flag hard-constraint violations.
4. Identify meaningful matches.
5. Explain important trade-offs.
6. Preserve uncertainty instead of inventing facts.
7. Learn useful patterns from user decisions.
8. Ask for confirmation when inferred preferences become consequential.
9. Improve future recommendations based on confirmed preferences.

Core principle: A job is not valuable because it scores highly. It is valuable when HeeraMon can explain why it fits the user’s priorities and whether the remaining trade-offs are worth considering.

12 Document Control

Version	Date	Author / Role	Changes
1.0	September 30, 2026	Product / Engineering	Initial release

Approval: This document defines the matching and preference behavior for HeeraMon. All subsequent design and implementation work must remain consistent with the principles stated herein.

HeeraMon flies to help people find job that fits not only their career, but the life they want to live.



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